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Course: Microeconomics
Assignment #6
Due Date: 12/6/2024

Booth Honor Code Pledge
We pledge our honor that we have adhered to the Booth Honor Code during this assignment.

X Aaron Bennett Date 12/5/2024

Problem 9 Suppose that the demand and supply for coffee are given by equations

$$Q^d = 1200 - 200p$$

$$Q^s = 200p.$$

(a). Find the equilibrium price and quantity for this market.

(b). Suppose that the government decides to buy up as much coffee as it takes to raise the market price to $p = 4$. Furthermore, suppose the government destroys all of its purchased coffee. How much deadweight loss does this intervention generate?

(c). Suppose instead that the government decides to subsidize coffee production by paying producers $1/bushel for every bushel of coffee sold. What are the after-subsidy equilibrium prices that the buyers effectively pay, p_b^*, and that the sellers effectively receive, p_s^*?

a) Solve by setting Qd = Qs

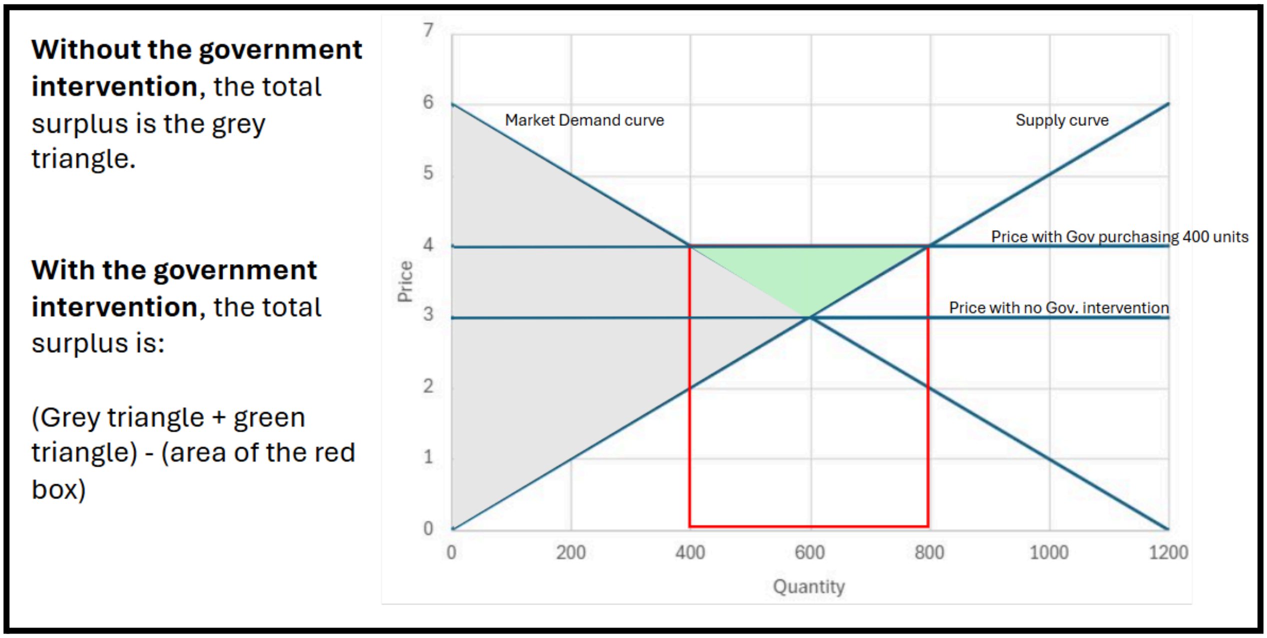
1200 - 200P = 200P**

*1200 = 400P**
P = 3*

Q = 3*200 = 600*

b) At a price of 4, suppliers are willing to supply 800 units (200 x 4), but consumers only want to purchase 400, so t**he government must buy the 400 units** that the consumers don't want to buy but that the suppliers are willing to supply.

The chart below shows how were were thinking about deadweight loss:



Since DWL = (Total Surplus without intervention) - (Total surplus after the intervention), the total DWL is $1,400.

One way to conceptualize this is that the government spends $1,600 to drive the price up from $3 to $4, but that action only generates an additional $200 of surplus. Since it took $1,600 to generate only $200 in surplus, then the DWL is $1,400 (the difference between those two values)

c) 1200 - 200Pb = 200Ps

1200 - 200Pb = 200 (Pb + 1)

1200 - 200Pb = 200Pb + 200

1000 = 400Pb

Pb* = 2.5

Ps* = Pb* + 1 = 3.5

Q = 200*Ps* = 700

Problem 11 True, False, Uncertain? [Explain your answer.] Suppose that demand for olives is elastic and the government decides to encourage production by offering olive growers a subsidy for each kilo of olives sold at market. *Total market expenditures by consumers on olives decrease following the subsidy.*

False

The government subsidy shifts the supply curve down. Because demand is elastic, the percent increase in quantity will be larger than the percent decrease in price, and revenue will increase.

Problem 16 Suppose that a monopolist's *opportunity* costs are given by the function $C(q) = 16 + 4q$ and the monopolist's demand is given by $q^d = 40 - 2p$. What is the firm's optimal choice of p and q?; what are the corresponding profits?

qd = 40 - 2P

2P = 40 - qd

P = 20 - qd/2

MR = 20 - qd

MC = dC/dq = 4

Setting MR = MC, we find: 20 - qd = 4

Therefore, qd* = 16

Pluggining back into the demand equation, when qd* = 16, p* = 12

Profit = Revenue - Cost = qd* x p* - C = 12 x 16 - (16 + 4 x 16) = $112